TalentAI: Multi-Agent Intelligent Recruitment System

System Architecture, Design Document & Input/Output Specification

Course: Continuous Assessment 3 (AI CA3) | Deliverables: Project Synopsis, Design Document, Input Definition, Output Definition | Status: Production Ready

1. Project Synopsis

The Problem: Modern recruitment pipelines and traditional Applicant Tracking Systems (ATS) are fundamentally hindered by high volumes of unstructured candidate documentation. Standard keyword-based ATS filtering suffers from catastrophic error modes: high false-positive rates caused by unqualified candidates 'stuffing' resumes with buzzwords, and high false-negative rates where qualified applicants are discarded simply for using alternative phrasing or synonyms. Furthermore, human talent acquisition specialists experience severe cognitive fatigue when screening hundreds of multi-page resumes against rapidly evolving technical criteria, introducing subjective bias, delays, and inconsistent qualification bars.

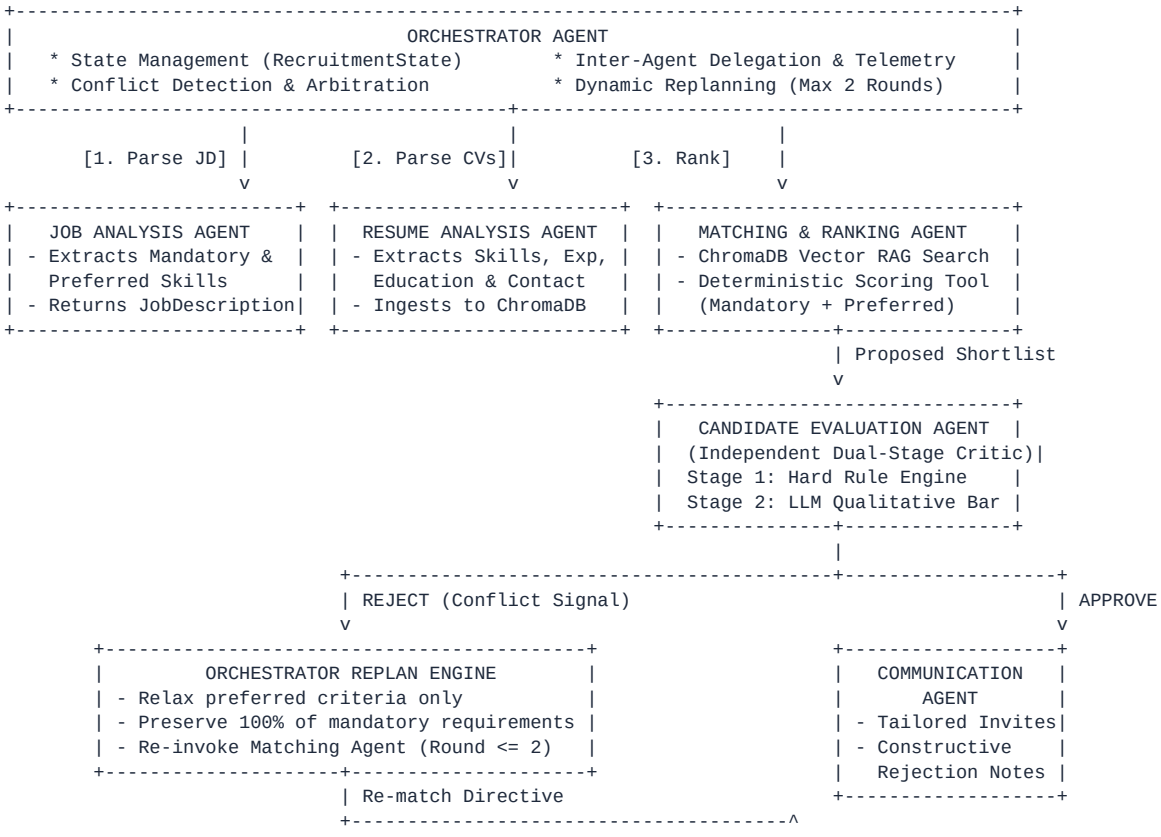
Our Solution: TalentAI is an autonomous multi-agent recruitment and candidate ranking platform that transforms technical hiring into a transparent, deterministic, and agentic workflow. Rather than relying on a single fallible prompt, TalentAI partitions the hiring pipeline across six specialized agents: Orchestrator, Job Analysis Agent, Resume Analysis Agent, Matching & Ranking Agent, Candidate Evaluation Agent (Dual-Stage Hybrid Critic), and Recruitment Communication Agent. The system pairs Retrieval-Augmented Generation (RAG) using dense vector embeddings (sentence-transformers *all-MiniLM-L6-v2*) in ChromaDB with deterministic requirement verification. Most critically, TalentAI demonstrates authentic agentic behavior via inter-agent conflict resolution: when the independent Evaluation Critic identifies disqualifying skill deficiencies in a proposed shortlist, the Orchestrator initiates dynamic replanning -- selectively relaxing secondary preferred qualifications while strictly enforcing 100% of non-negotiable mandatory skills.

Target Users & Scope Evolution (matching approved CA2 project description): The platform is tailored for corporate talent acquisition teams, technical recruiters, engineering hiring managers, and small-to-medium enterprises requiring rigorous first-round technical screening.
Scope Evolution Note: In the approved CA2 project proposal, the system was originally planned as a linear single-pass retrieval pipeline with heuristic scoring. During CA3 implementation, the architecture was intentionally expanded to resolve reviewer feedback regarding naive keyword matching: (1) we introduced an independent **Dual-Stage Hybrid Critic** in the Evaluation Agent combining deterministic python verification with LLM qualitative auditing; (2) we implemented **Dynamic Replanning and Conflict Reconciliation** in the central Orchestrator with bounded iterations to prevent infinite loops; and (3) we developed a real-time telemetry streaming interface via FastAPI Server-Sent Events (SSE) and Next.js 14 to provide transparent audit trails of multi-agent deliberation.

2. Design Document

2.1 System Architecture & Agent Communication Topology

The architecture implements a centralized **Orchestrator-Worker** topology with an independent quality evaluation gate. Inter-agent communication flows deterministically through the central Orchestrator, ensuring state consistency and bounded execution.



2.2 Agent Roles and Responsibilities

Agent	Role & Responsibilities	Input / Output	Tools & Tech
1. Orchestrator Agent	Central controller. Delegates pipeline stages, monitors inter-agent outputs, handles state transitions, reconciles Evaluation conflicts, and triggers dynamic replanning with bounded iterations (max 2).	**In:** RecruitmentRequest **Out:** RecruitmentState, SSE Stream	CrewAI Agent, State Machine, Groq LLaMA-3.3-70B
2. Job Analysis Agent	Parses raw job descriptions to isolate non-negotiable mandatory skills, preferred qualifications, experience constraints, role title, and company profile into structured format.	**In:** Raw JD string **Out:** JobDescription Model	Groq LLaMA-3.3-70B (JSON entity extraction)
3. Resume Analysis Agent	Parses candidate resumes in parallel. Extracts structured contact info, verified technical skills, professional vs. internship experience, and indexes document embeddings into vector space.	**In:** Raw Resumes list **Out:** Candidate Model list	Groq LLaMA-3.1-8B-Instant, ChromaDB Ingestion
4. Matching & Ranking Agent	Executes semantic vector search across candidate embeddings and applies composite scoring formula (Mandatory 60%, Preferred 30%, Experience 10%). Supports relaxed-mode rerun upon replan.	**In:** JobDescription, Candidates **Out:** MatchedCandidate list	RAGRetrievalTool (ChromaDB), ScoringTool
5. Candidate Evaluation Agent	Independent quality gate. Implements a Dual-Stage Hybrid Critic: Stage 1 deterministically checks 100% mandatory compliance; Stage 2 performs qualitative audit of engineering depth. Emits APPROVE or REJECT.	**In:** Shortlist, JobDescription **Out:** Verdict, Audit Notes	Independent LLM Call, Deterministic Rule Engine
6. Communication Agent	Generates personalized, professional interview invitations highlighting specific achievements for approved candidates, and respectful, constructive feedback notices for non-selected applicants.	**In:** Evaluated Shortlist **Out:** Formatted Outreach Texts	Groq LLaMA-3.3-70B (Generative NLG)

2.3 Orchestration Pattern: Centralized Orchestrator-Worker with Conflict Loop

TalentAI employs a **Centralized Orchestrator-Worker pattern** with an autonomous conflict resolution feedback loop. Direct peer-to-peer agent messaging was deliberately avoided to eliminate circular deliberation deadlocks and ensure that state updates remain atomic and deterministic.

Conflict Resolution Protocol:

- **Initial Proposed Shortlist:** The Matching Agent produces a top-N candidate shortlist based on semantic similarity and composite scoring.
- **Adversarial Evaluation Gate:** The Candidate Evaluation Agent acts as an adversarial quality gate (critic). If a candidate possesses high semantic similarity but lacks mandatory skills (e.g., Arjun Mehta matching React/Node but lacking TypeScript and PostgreSQL), the Evaluation Agent issues an explicit **REJECT** conflict signal.
- **Strategic Replanning Arbitration:** The Orchestrator receives the rejection and executes orchestrator_decide_replan(), selecting from four actions: replan_relaxed, replan_strict, escalate, or accept_partial.
- **Strict Mandatory Preservation Rule:** Under this rule, the Orchestrator relaxes secondary preferred criteria (e.g. AWS, GraphQL) while holding mandatory requirements 100% inviolable, instructing the Matching Agent to search for qualified alternative candidates.
- **Loop Boundedness:** Re-matching occurs with a hard ceiling of **MAX_REPLAN_ROUNDS = 2** to guarantee deterministic termination.

2.4 Technology Choices and Architectural Justifications

Component	Technology Selected	Version / Spec	Engineering Justification
Agent Framework	CrewAI	1.15.x	Provides native Agent, Task, and Crew abstractions; clean tool binding; native support for structured output schemas and role-based execution.
Primary LLM	Groq LLaMA-3.3-70B	70B Versatile	Sub-second inference (<800ms) on Groq LPUs avoids HTTP timeouts during multi-agent turns. Superior reasoning for JD extraction and qualitative auditing.
Fast Parser LLM	Groq LLaMA-3.1-8B	8B Instant	High-throughput, cost-effective extraction for bulk resume ingestion without sacrificing entity extraction accuracy.
Vector Database	ChromaDB	0.5.x / 1.x	Embedded local vector store with zero cloud latency, zero external API costs, local persistence, and native cosine similarity querying.
Embedding Model	sentence-transformers	all-MiniLM-L6-v2 (384-d)	Lightweight, runs locally on CPU/GPU, trained on 1B+ sentence pairs, captures semantic nuances between resume terminology and job specifications.
Backend API	FastAPI	0.115.x	Asynchronous event loop, native Server-Sent Events (SSE) streaming for real-time agent telemetry, and native Pydantic schema validation.
Frontend UI	Next.js 14	App Router / TS	Modern reactive interface, real-time SSE stream consumption, interactive candidate dossiers, and live multi-agent deliberation telemetry feed.

3. Input Definition

The TalentAI system accepts recruitment requests through the REST and SSE streaming endpoints (POST /api/recruit and POST /api/recruit/stream). The input format is strictly validated via Pydantic.

3.1 Input Schema (RecruitmentRequest)

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "RecruitmentRequest",
  "type": "object",
  "properties": {
    "job_description": {
      "type": "string",
      "description": "Raw unformatted text of the job description (minimum 50 characters)."
    },
    "resumes": {
      "type": "array",
      "items": { "type": "string" },
      "description": "List of raw unformatted candidate resume texts (minimum 1 resume)."
    },
    "top_n": {
      "type": "integer",
      "default": 5,
      "minimum": 1,
      "description": "Target number of top-matching candidates to shortlist."
    }
  },
  "required": ["job_description", "resumes"]
}
```

3.2 Real Sample Input (Production Scenario)

```
POST /api/recruit/stream HTTP/1.1
Host: 127.0.0.1:8000
Content-Type: application/json

{
  "job_description": "Senior Full Stack Engineer at TechNova Inc.\nWe are seeking a Senior Full Stack Engineer with 4+ years experie
  "resumes": [
    "Kavya Reddy | kavya.reddy@email.com | 7 years exp\nSKILLS: TypeScript, React, Node.js, PostgreSQL, Docker, Kubernetes (CKA), AW
    "Priya Sharma | priya.sharma@email.com | 6 years exp\nSKILLS: React, Node.js, TypeScript, PostgreSQL, Docker, AWS (EC2, S3, RDS)
    "Arjun Mehta | arjun.mehta@email.com | 3 years exp\nSKILLS: React, JavaScript, Node.js, Express, MySQL, basic PostgreSQL, Git, A
    "Aisha Patel | aisha.patel@email.com | 5 years exp\nSKILLS: Python, FastAPI, PostgreSQL, AWS, Docker, Kubernetes, Microservices,
  ],
  "top_n": 3
}
```

4. Output Definition

TalentAI produces two complementary output formats: (1) a real-time event-driven **Server-Sent Events (SSE) telemetry stream** detailing each agent's decisions and conflict occurrences, and (2) a complete **RecruitmentResponse** containing the final recruitment state, candidate dossiers, eligibility audits, replan history, and generated communication drafts.

4.1 Output Schema (RecruitmentResponse)

```
{
  "title": "RecruitmentResponse",
  "type": "object",
  "properties": {
    "state": {
      "type": "object",
      "properties": {
        "job": { "$ref": "#/definitions/JobDescription" },
        "candidates": { "type": "array", "items": { "$ref": "#/definitions/Candidate" } },
        "shortlisted": { "type": "array", "items": { "$ref": "#/definitions/MatchedCandidate" } },
        "evaluation_passed": { "type": "boolean" },
        "replan_count": { "type": "integer" },
        "relaxed_requirements": { "type": "boolean" },
        "final_communications": { "type": "array", "items": { "type": "string" } },
        "status": { "type": "string", "enum": ["pending", "running", "completed", "failed"] },
        "final_status": { "type": "string", "enum": ["COMPLETED_SUCCESS", "COMPLETED_PARTIAL", "COMPLETED_NO_MATCH"] }
      }
    },
    "execution_trace": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "agent": { "type": "string" },
          "role": { "type": "string", "enum": ["action", "decision", "conflict", "replan", "result"] },
          "content": { "type": "string" },
          "timestamp": { "type": "string", "format": "date-time" }
        }
      }
    }
  }
}
```

4.2 Real Sample Output (Resolved Multi-Agent Execution)


```
{
  "state": {
    "job": {
      "id": "job-tech-01",
      "title": "Senior Full Stack Engineer",
      "company": "TechNova Inc.",
      "mandatory_requirements": ["react", "node.js", "typescript", "postgresql", "docker", "kubernetes", "aws"],
      "preferred_requirements": ["graphql", "redis", "microservices"],
      "min_experience_years": 4.0
    },
    "shortlisted": [
      {
        "candidate": { "name": "Kavya Reddy", "email": "kavya.reddy@email.com", "experience_years": 7.0 },
        "match_score": 90.0,
        "matched_mandatory": ["react", "node.js", "typescript", "postgresql", "docker", "kubernetes", "aws"],
        "missing_mandatory": [],
        "matched_preferred": ["graphql", "microservices"],
        "eligibility_status": "FULLY_ELIGIBLE",
        "evaluation_notes": "Exceeds all criteria. CKA certified Kubernetes + AWS Architect. APPROVED by EvaluationAgent."
      },
      {
        "candidate": { "name": "Priya Sharma", "email": "priya.sharma@email.com", "experience_years": 6.0 },
        "match_score": 85.0,
        "matched_mandatory": ["react", "node.js", "typescript", "postgresql", "docker", "aws"],
        "missing_mandatory": [],
        "matched_preferred": ["microservices"],
        "eligibility_status": "FULLY_ELIGIBLE",
        "evaluation_notes": "Solid enterprise full-stack engineer. Meets 100% of mandatory bar. APPROVED."
      },
      {
        "candidate": { "name": "Aisha Patel", "email": "aisha.patel@email.com", "experience_years": 5.0 },
        "match_score": 72.0,
        "matched_mandatory": ["postgresql", "docker", "kubernetes", "aws"],
        "missing_mandatory": [],
        "matched_preferred": ["microservices"],
        "eligibility_status": "CONDITIONALLY_ELIGIBLE",
        "evaluation_notes": "Surfaced in Round 2 Replan after Orchestrator relaxed preferred criteria. Replaced rejected candidate A
      }
    ],
    "evaluation_passed": true,
    "replan_count": 1,
    "relaxed_requirements": true,
    "status": "completed",
    "final_status": "COMPLETED_SUCCESS",
    "final_communications": [
      "Dear Kavya, We were thoroughly impressed by your 7 years of full-stack engineering and Kubernetes expertise at Razorpay. Tech
      "Dear Priya, Your 6 years of scalable React/Node.js enterprise development at Infosys stood out as an exceptional fit. We woul
      "Dear Aisha, Your extensive distributed backend systems experience at Google India is an exciting match. We would love to disc
    ]
  },
  "execution_trace": [
    { "agent": "Orchestrator", "role": "action", "content": "Pipeline initialized for 4 candidates.", "timestamp": "2026-10-21T18:40
    { "agent": "JobAnalysisAgent", "role": "action", "content": "Extracted 7 mandatory and 3 preferred requirements.", "timestamp":
    { "agent": "MatchingAgent", "role": "action", "content": "Ranked candidates: Proposed Kavya (87.5), Priya (82.0), Arjun (41.0).'
    { "agent": "EvaluationAgent", "role": "conflict", "content": "CONFLICT: Shortlist REJECTED. Arjun Mehta critically lacks mandato
    { "agent": "Orchestrator", "role": "replan", "content": "Dynamic Replan initiated: REPLAN_RELAXED. Relaxing secondary preferred
    { "agent": "MatchingAgent", "role": "replan", "content": "Re-ranking complete. Aisha Patel surfaced (72.0). Arjun Mehta removed.
    { "agent": "EvaluationAgent", "role": "result", "content": "VERIFIED & APPROVED. Revised shortlist meets all mandatory qualifica
    { "agent": "CommunicationAgent", "role": "result", "content": "Bespoke interview invitation letters generated.", "timestamp": "2
  ]
}
```